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Genetically and semantically aware homogeneous network for prediction and scoring of comorbidities

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ABSTRACT

Objective: Patients with comorbidities are highly prone to mortality risk than those suffering from a single disease. Therefore, quantification and prediction of disease comorbidities is necessary to stratify the mortality risk of the patients, predict the probability of their occurrence, design treatment strategies, and to prevent the progression of diseases. Enriching comorbidity disease relationships with rich semantics established by genetic components play a vital role in effectively quantifying and predicting comorbidities. However, the existing studies have not extensively explored the semantic richness conveyed by different types of genetic links connecting the comorbidity pairs.
Methods: To solve this, a novel genetic-semantic aware weighted homogeneous network-based method, GSWHomoNet is proposed which first constructs the gene enriched comorbidity heterogeneous network, CoG-HetNet with encoded genetic semantic aware weighted meta-path instance disease pair embedding to obtain an enhanced disease node embedding of the network. For enhanced comorbidity prediction and scoring, both direct and indirect semantically enriched comorbidity relationships of the disease nodes is preserved while transforming heterogeneous to homogeneous comorbidity network GSWHomoNet. The proposed GSWHomoNet not only helps discover comorbidity links transductively between known-known disease pairs but also improves the inductive link prediction between known-unknown disease pairs by supplying unknown disease nodes with semantically enriched heterogeneous structural knowledge.
Results: The effectiveness of the proposed components is proved by AUC scores of 0.895 and 0.860, as well as AUPR scores of 0.903 and 0.873 for transductive and inductive link prediction respectively. In comorbidity scoring, GSWHomoNet outperformed other methods with a correlation result of 0.848. The effect of the improved association prediction ability of the genetic semantic aware weighted meta-path instance embedding based node embedding is proved on disease-microbe and bibliographic heterogeneous network datasets. For biological significance of GSWHomoNet-based comorbidity scoring, we compared it with gene, pathway, and protein-protein interaction (PPI) perspectives, revealing a stronger correlation with the PPI aspect. We identified a substantial number of predicted comorbidity disease pairs, with 77,456 and 48,972 pairs supported by literature evidence for transductive and inductive predictions, respectively. Additionally, we highlighted shared pathways and PPIs for these pairs, demonstrating the robustness of comorbidity predictions.

1. Introduction

Disease comorbidity is defined as two or more diseases that co-exist in the same patient simultaneously [1,2] whose effect includes adverse health outcomes, prolonged hospital stays, complex treatment management and increased probability of being mortal [3]. Existing treatment plans are designed for individual diseases, posing challenges in effectively managing patients with comorbidities. Therefore, an effective quantification and prediction of comorbidities can facilitate efficient management and treatment plans. Estimating the probability of occurrence of a disease due to another in the same individual is given by comorbidity scoring [4] which stratifies patients into low, medium, and high-risk survival types and further helps to break the comorbidity disease chain [5]. In addition, prediction of the comorbidities can help early diagnosis and prevention of diseases that have a higher probability of occurrence over time.

Two diseases are associated because of sharing and non-sharing of many genetic (gene ontology (GO) function, protein-protein interaction

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(PPI), pathways, genetic variant Single Nucleotide Polymorphism (SNP), micro–Ribonucleic Acid (miRNA), gene) and non-genetic biological components (drugs, symptoms, metabolites, patient clinical information, environmental factors) [6–10] and factors such as ontological connections like specificity of gene ontology functions associated to the gene sets of the diseases [11] and other hierarchical-based approaches using disease ontology [12]. Text-based co-occurrences like two diseases appearing in the same sentence in biomedical literature also contribute to the disease-disease association [13–15]. Comorbidities are subsets of disease-disease associations and exists as a result of one disease causing another due to the alterations of genes in individuals. Therefore, the genetic aspects of diseases or the disease related genetic components such as Gene Ontology (GO) function [16], PPI [17], pathway [17,20], genetic variant (SNP) [18], miRNA [19,20] along with gene play a crucial role in comorbidities. The specific entities of each genetic component are the genetic instances. For example, the genetic component *gene* has genetic instances *KLF14, ADAMTS9, FTO, HHEX* etc. associated with disease *diabetes* [8]. Exploring these genetic oriented disease associations is essential to understand the pathogenesis of comorbidity and for detecting the comorbid pairs and scoring them. For better comorbidity quantification and comorbidity disease link prediction (predicting the likelihood of comorbid links between disease nodes in a network), a deeper understanding of comorbidity semantic context, that is, various meaningful genetic connections between comorbid diseases are necessary.

Most of the existing work used statistical approaches to quantify and predict disease comorbidities [2,16,17,20]. Such statistical characterization of comorbidity is based only on quantitative genetic information contributed by only few independent genetic components and their genetic instances associated with diseases [21]. Thus, the existing statistical comorbidity approaches incorporated only limited aspects/context of comorbidity without linking the genetic components.

In order to fully exploit the context of comorbidities, interconnections between disease-associated genetic components can be analyzed using biological heterogeneous networks by employing the network-based approaches for comorbidity link prediction and scoring [19,22,23]. In general, heterogeneous networks consist of multiple types of nodes and relationships or edges. By using heterogeneity information, Shuting Jin et al. [19] have explored the genetic biological components such as gene and miRNA while genes, PPI, GOfunction, pathway have been explored by Biswas et al. [22]. Another work [23] used homogeneous disease network, where the link between diseases was established by using only SNP. However, a broader comorbidity context can be captured well by interconnecting a variety of biological components from massive biological data sources and constructing a heterogeneous biological network, which is the initial crucial step for carrying out comorbidity task. Some studies embedded the interconnected biological components with features in the biological network. For microbe-disease association prediction, Yali Chen et al. [24] constructed a microbe-drug-disease network with embedded node features. For comorbidity relation prediction, Biswas et al. [22] embedded the features of various genetic components such as disease, gene, GO function, pathway and integrated them to build a relational heterogeneous network or knowledge graph. However, the embedded features of the components are independent of the various gene aspects of comorbidity. Thus, the existing network-based methods used only limited genetic components and captured limited comorbidity context.

Further, various network information (structural and semantic relations [25,26]) in the constructed biological heterogeneous network [27–31] is learnt using the network embedding approaches such as graph neural network (GNN) and meta-path-based methods. In general, a meta-path is defined as a path with sequence of node types starting from original node to destination node. Some work leveraged GNN to generate node embeddings such as Variational Graph Attention Encoder (VGAE) and GraphSAGE were employed together on a heterogeneous network consisting of pathways, drugs, SARS-CoV-2 to produce drug embeddings which were used to select the most candidate drugs for the disease COVID-19 [32]. To predict disease for patients with comorbidities, the patient node embeddings were learnt using Graph Convolutional Network (GCN) and Graph Attention Network (GAT) by aggregating the neighbor patient node information from a patient network constructed using administrative claims data [33]. Using multiple similarity graphs, miRNA-disease and microbe-disease association predictions were carried out by generating their embeddings using Multi-view Multichannel Graph Convolutional Network (MMGCN) [34] and GAT-based framework [35] respectively. Though GNN models have been applied to generate node embeddings in biomedical tasks, they mainly fuse the neighboring node information and may not fully capture the complex patterns in heterogeneous graph. Meta-path based methods can efficiently handle heterogeneous network with multiple types of nodes and relationships and allow for direct integration of domain knowledge by defining meaningful paths. Therefore, studies related to meta-path based node embedding are discussed in detail. Some efforts have been put-forth to generate node representation by embedding the meta-path instances which incorporate the latent characteristics of all nodes along the meta-path instance. In this regard, a node embedding was carried out by GraphMSE [36] using meta-path based aggregation of semantic information from multiple meta-path instance embeddings where all meta-path instances under a meta-path were assumed to possess the same semantics. Meta-path instance embedding was obtained by concatenating latent embedding of the intermediate nodes. On the other hand, some of the meta-path aggregated node embedding methods such as MAGNN [51] and MATHDMA [24] for disease-microbe association prediction derived node representation by aggregating multiple attention-based meta-path embeddings. The semantic information of each meta-path was obtained by aggregating multiple meta-path instance embeddings conveying only one semantic type. Further, the meta-path instance embedding was generated by aggregating the intermediate node embeddings without considering the ordered dependencies among the intermediate nodes. However, an attempt was made to maintain the order of the nodes using LSTM cells. In this regard, Xiang Wang et al. [37] utilized the LSTM cells and developed a model KPRN to exploit the knowledge graph (KG) in order to provide reasoning for recommendation by learning user-item pair based different meta-path instance representations using latent features of nodes and different edge relations.

Though the above methods differentiated the semantics among different meta-path instances, they failed to consider the contribution effect of nodes as well as component edges along the meta-path instance during meta-path instance embedding. Another major issue is that they did not consider the length and count of various path instances connecting the pair. Thus, the existing meta-path-based node embedding methods failed to differentiate the deeper semantics of meta-path instances connecting the node pairs. Considering the importance of nodes is required as different nodes show different contributing degree to the meta-path instance pair connection. In addition, considering the importance of component edges is essential as the same component edge may participate in multiple meta-path instances whose relative importance varies with respect to the pair. Moreover, both the number and varying length of meta-path instances between node pairs convey stronger or weaker semantic strength between the pair. From the comorbidity perspective, there is a need to fully exploit the meta-path instance information as it affects the quality of final node representation which is significantly important for comorbidity prediction and scoring.

In addition to the heterogeneous network information, the performance of comorbidity quantification and prediction could be improved with homogeneous network information. Given a heterogeneous network, some methods used the heterogeneous network information to transform heterogeneous to homogeneous network. Using heterogeneous meta-path decomposition for recommendation, HERec [38] applied DeepWalk while AMERec [39] and H2Rec [40] applied the random walk strategy to multiple decomposed meta-path based homogeneous networks. While transforming heterogeneous to homogeneous network, HERec and AMERec failed to consider multiple semantics conveyed by intermediate nodes of multiple meta-path instances leading to semantic information loss. Though H2Rec [40] considered the intermediate nodes and learnt meta-path based representation using multiple meta-path instance sequence of nodes, they did not consider the deeper semantic information of nodes and edges. Moreover, the models considered two nodes to be connected in the transformed homogeneous network without considering the length and number of multiple meta-path instances connecting them. However, AMERec weighted the edges in the transformed homogeneous network, calculated using only the count of meta-path instances which were independent of the semantic information conveyed by the heterogeneous network.

While transforming heterogeneous to homogeneous network, the existing methods failed to incorporate the heterogeneous semantic richness of nodes and links in the transformed homogeneous network. Moreover, the length and number of multiple meta-path instances connecting the node pairs were independent of the semantic information conveyed by the heterogeneous network [39].

Further, the link prediction can be carried out by learning various kinds of network information [15,41–43] including the transformed homogeneous network using heterogeneous information of the pairs to discover new pair link. Link prediction can be classified into two types: 1) transductive link prediction that predicts new links between known nodes visible to the model in the training phase 2) inductive link prediction which occurs between new or unknown and known nodes in the network where the model is unaware of the new node during training. In general, the comorbidity links are discovered between known disease nodes [19,22,44] using the fact that only disease nodes exist in the network are embedded with comorbidity context. However, these methods fail to predict new comorbidity link when a new disease emerges (unknown node) that is not present in the network. Such kind of link prediction is addressed as inductive or cold start link prediction [45]. For this purpose, inductive link prediction methods DEAL [45] and LeSICiN [46] used an alignment mechanism to align both node's attribute as well as structural information during model training. During testing, when a new node arrives, it has only attribute information and no structural information for the trained model to further predict links. In a semi-connected network consisting of the isolated nodes, Minghu Tang and Wenjin Wang [47] used additional structural information of those isolated/new nodes using community structure derived from clustering of similar attribute features of nodes. Thus, only the attributes similarity-based structural information of new nodes is exploited disregarding the actual topological network features for inductive link prediction [45,46]. The rich heterogeneous indirect semantic connectivity information of the unknown node with other homogeneous nodes was not exploited. Therefore, for improving the performance of unknown node's link prediction in a comorbidity homogeneous network, there is a need to supply the unknown node with structural and semantic comorbidity heterogeneous connectivity information with other nodes.

Overall, the above discussed studies, have not exploited multiple biological components with richer node features. The meta-path based node embedding requires a detailed consideration of semantic information of meta-path instances. While transforming heterogeneous to homogeneous network, the richer heterogeneous network information must be preserved in the transformed homogeneous network.

In this work, a novel genetic-semantic aware weighted homogenous network-based method, GSWHomoNet is developed as illustrated in Fig. 1, for better comorbidity link prediction and scoring with several major contributions as follows.

1. Enriched and interconnected the various gene aspects of several genetic componenttypes associated with comorbidity such as gene, GO, pathways, miRNA, SNPs and PPIs.
2. To obtain an enriched disease node embedding, we proposed a rich informative genetic-semantic aware weighted meta-path instance embedding method that emphasizes the significance of considering each meta-path instance semantically different.
3. We learnt both direct and semantically enriched indirect disease relationship in a semantically embedded and weighted disease homogeneous network obtained from the constructed heterogeneous network. This is to carry out comorbidity transductive and inductive link prediction and scoring using the homogeneous network.

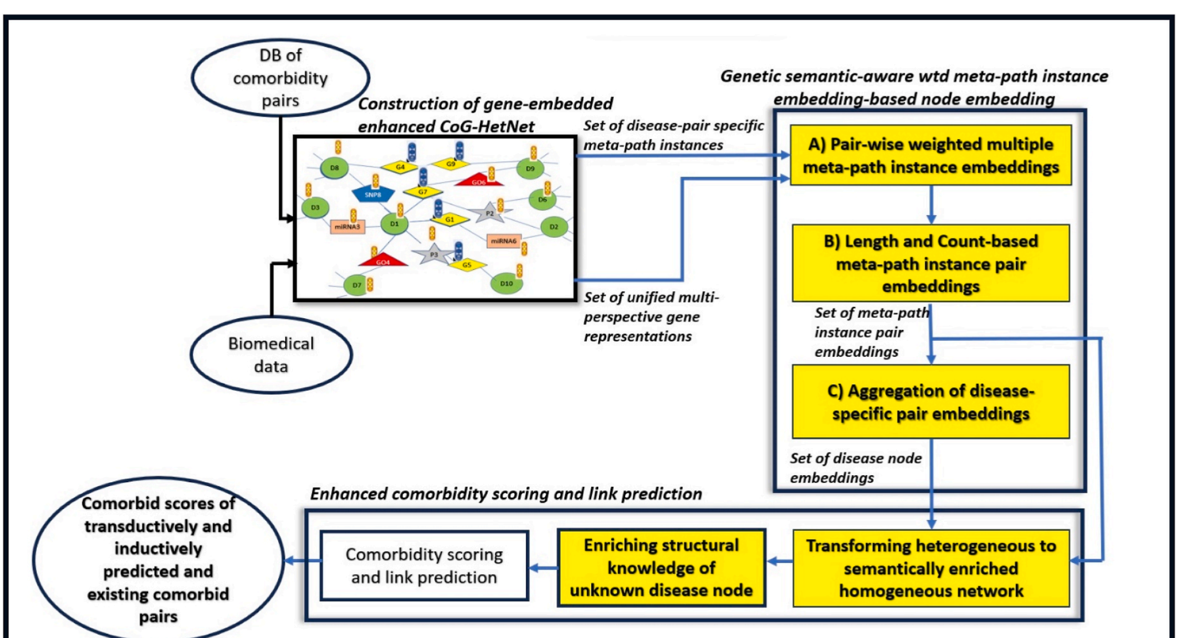

Fig. 1. Overview of genetic-semantic aware weighted homogeneous network-based method, GSWHomoNet.

2. Materials and methods

In this study, a novel *genetic-semantic aware weighted homogeneous network-based method, GSWHomoNet* was developed as shown in Fig. 1, to effectively quantify and predict comorbidities using both genetically and semantically rich heterogeneous and homogeneous network information. The developed method has three major modules: (i) *Construction of gene embedded enhanced CoG-HetNet* using the available comorbidity pairs and the biological components and their associations from biomedical data. (ii) Obtaining *Genetic-semantic aware weighted meta-path instance embedding based disease node embedding* using the extracted disease-pair meta-path instances and the enhanced gene representation of the nodes from CoG-HetNet (iii) *Enhanced comorbidity link prediction and scoring* by transforming heterogeneous to semantically enriched weighted homogeneous network using the meta-path instance pair embeddings and disease node embeddings from (ii) and by enriching the structural knowledge of unknown disease node. Finally, the transductively and inductively predicted comorbid pairs along with the existing comorbid pairs are quantified to derive the comorbidity scores.

We next describe the definitions in section 3.1 and the notations in Table 1 which are necessary to describe the proposed work.

2.1. Preliminary

In this section, we introduce some terminologies related to heterogeneous biological networks. Besides, the summary of the notations used in this paper is given in Table 1 for a quick look-up.

1) *Heterogeneous network:* A biological heterogeneous network is shown in Fig. 2 which consists of 5 node types (Disease D, Gene G, Gene Ontology function GO, Genetic variant SNP and miRNA) with different association types connecting each other. Some of the network properties include sparsity, centrality of nodes, network size, density, degree distribution, path length, clustering coefficient, diameter of network, connectedness, community detection etc.

Table 1
Notations used in this paper.

Notations	Definitions	Notations	Definitions
G	Gene	W, b	Weight matrix, bias
G^{PPI}	PPI-based gene representation learning (RL)	h_{n_k}	Genetic embedding of node n_k
G^{BP}, G^{GE}	biological properties-based, gene expression vector representation (VR)	. $h_{n_k}^{GC_i}$	Embedding of node n_k influenced by a specific genetic component GC_i
$< G >$	aggregated set of unified multi-perspective gene representations	h_{GC_{ij}/n_k}	Local genetic-instance GC_{ij} embedding w.r.t node n_k
n_k	node in CoG-HetNet	$\alpha_{n_k}^{<GC_i>}$	Genetic component GC_i-based node n_k attention
D	Disease	w	Component edge weight
$D_x - D_y$,	Disease pair with diseases D_x *and* D_y	mp_k	Meta-path instance k
GC	Genetic component	l	Length
GC_{ij}	Genetic instance j of ith genetic component	MLP_l	Length-wise multi-layer perceptron
X	Any genetic component such as G, GO, pathway, SNP, miRNA	$path\_count^l$	# of meta-path instances of length l
α, β	Attention weights	$h(D_x - D_y)$	Disease pair embedding
h	Learnt embedding vector	$h(D_x)$	$D_x's$ embedding
V	Set of all nodes in CoG-HetNet	$SG_{D_i,D_j}^{K \times L}$	Sub-graph pair with sub-graphs $SG_{D_i \to D_j}^{K}$ and $SG_{D_j \to D_i}^{L}$

2) *Meta-path:* In Fig. 2, two disease nodes (D-D) are connected by multiple meta-paths consisting of different types of genetic components such as G, GO, SNP, miRNA. Each meta-path conveys different semantics of connection between diseases. For example, $D - G - miRNA - D$ represents a semantic connection between diseases in the order of $D - G, G - miRNA$ and then $miRNA - D$ which is different from $D - miRNA - G - D$.
3) *Meta-path instances:* Given a meta-path, the disease and other genetic component nodes when substituted by disease instances D_1, D_2, D_3, … and different genetic instances can result in multiple meta-path instances that convey additional semantics. In other words, a meta-path can hold multiple semantics conveyed by multiple meta-path instances. The figure shows $D - G - miRNA - D$ meta-path consists of meta-path instances - $D1 - G1 - miRNA6 - D2$ and $D1 - G1 - miRNA7 - D2$, having different semantics contributed by different miRNA instances $miRNA6$ and $miRNA7$ respectively.

2.2. Construction of gene embedded enhanced CoG-HetNet

For the comorbidity heterogeneous network construction, it is essential to first collect the comorbidity disease pairs from Hudine data, http://sbi.upf.edu/data/hudine/. The attributes of the dataset are ICD9 5-digit level codes of disease 1 and disease 2, frequency of co-occurrence of disease pair (n.common), relative risk (RR) calculated by Hidalgo et al. [1], along with some additional attributes. We used a similar criterion [16] to retain disease pairs with strong indication of comorbidity. We then collected disease and gene associated genetic instances of various genetic components given in Table 2 and interconnected them to form a comorbidity perspective gene-based heterogeneous CoG-HetNet. As most of the disease-genetic association mappings uses MeSH identifiers, we mapped a total of 1242 ICD9 diseases from Hudine data to MeSH disease identifiers using oxo ontology tool (https://www.ebi.ac.uk/spot/oxo/search#). The MeSH mapped ICD9 disease ids are used to get other genetic associations.

The following sections 3.2 to 3.4 introduce the proposed modules of the novel GSWHomoNet method.

Each node in CoG-HetNet is enriched by an *aggregated set of unified multi-perspective comorbidity-based gene representations of its associated genes*, $n_k^{<G>}$ obtained from various sources given in Table 2. An enhanced multi-perspective gene representation for each gene G is the unification of the embeddings and vector representations. The embeddings are learnt using topological properties of genes from PPI similar to Sezin et al. [48] and is denoted as PPI-based gene representation learning (RL) G_{PPI}. The vector representations include the biological properties-based vector representation (VR) of genes G_{BP} represented using one-hot encoding [48] and the gene expression vector representation (VR) G_{GE} using varying expression levels of genes contributing to diseases [49] taken from GEO [50]. Thus, the resulting network becomes a gene embedded enhanced CoG-HetNet enabling it to have a major impact on comorbidity prediction and scoring carried out in the subsequent modules. The enhanced CoG-HetNet can then be used to extract a set of unified multi-perspective gene representations of the nodes and set of disease-pair specific meta-path instances of lengths $l = 2,3$ and 4. For $l = 4$, we consider only meta-paths following pattern $D - X - G - X - D$, as each disease is indirectly connected to its genetic component X (GO, pathway, SNP, miRNA) via its associated gene G.

2.3. Genetic-semantic aware weighted meta-path instance embedding based disease node embedding

Using the set of disease pair-specific meta-path instances of lengths 2,3 and 4 and the unified set of multi-perspective gene representation of nodes from CoG-HetNet, a pair-wise weighted meta-path instance embeddings is created, which is used for generating length and count based meta-path instance pair embeddings and these are aggregated to get disease-specific pair embeddings to derive the final disease node

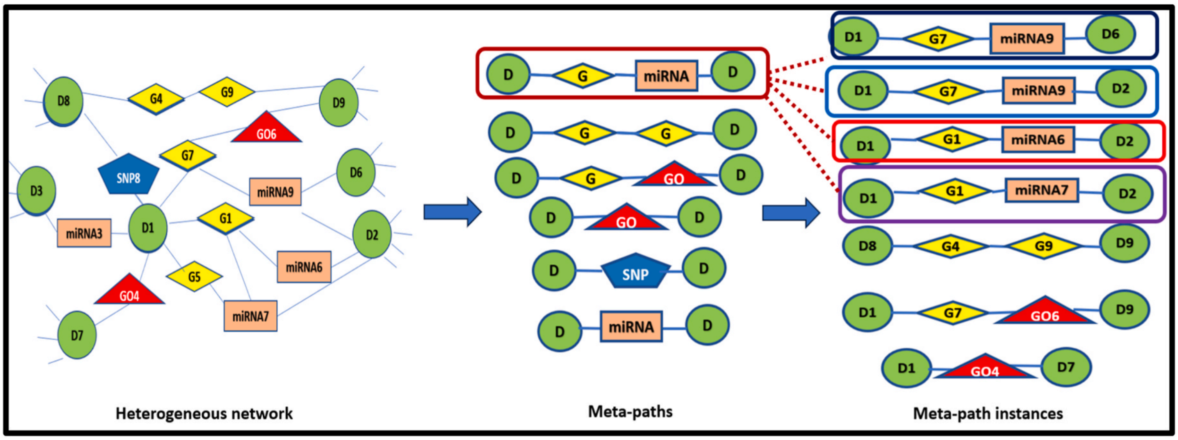

Fig. 2. Illustration of meta-path and meta-path instances.

Table 2
Statistics of heterogeneous enhanced gene-based comorbidity network CoG-HetNet (a) Nodes (b) Associations.
1) Local-neighborhood genetic component-based node attention

(a) Nodes

Nodes	Count	ID	Source database
Disease	1242.0	ICD9	Hudine data source
Gene	26237.0	Entrez ID	NCBI
GO Function	19615.0	Gene Ontology ID GO ID	Gene Ontology
Pathway	2294.0	KEGG & Reactome ID	CTD
SNP	4045.0	dbSNP ID	PheWas Catalog
miRNA	795.0	miRbase ID	miR2Disease & HMDV3.0

(b) Associations

Associations	Count	ID	Source database
Disease-Gene (D-G)	10351461.0	ICD9 mapped MESH ID-Entrez ID	CTD
Disease-GO_Func (D-GO)	1608145.0	ICD9 mapped MESH ID-GO ID	CTD
Disease-pathway (D-Pathway)	250903.0	ICD9 mapped MESH ID-KEGG/Reactome ID	CTD
Disease-SNP (D-SNP)	272166.0	ICD9 mapped MESH ID-dbSNP ID	PheWas Catalog
Disease-miRNA (D-miRNA)	15064.0	ICD9 mapped MESH ID- miRBase ID	miR2Disease & HMDV3.0
Gene-Gene (G-G) (PPI)	246536.0	Entrez ID-Entrez ID	Human PPI built by mpDisNet [19]
Gene-GO_Func (G-GO)	295811.0	Entrez ID-GO ID	NCBI gene2go
Gene-Pathway (G-Pathway)	28708.0	Entrez ID- KEGG/ Reactome ID	CTD
Gene-SNP (G-SN)	1618.0	Entrez ID-dbSNP ID	DisGeNET
Gene-miRNA (G-miRNA)	89291.0	Entrez ID-miRBase ID	miRTarBase

embedding as shown in Fig. 3. The obtained disease node embedding and weighted length and count based pair embeddings are further used for comorbidity link prediction and scoring.

2.3.1. *Pair-wise weighted meta-path instance embeddings*

Given a pair connected by multiple meta-path instances of varying lengths, it is necessary to learn different semantics of the pair using pair-wise weighted multiple meta-path instance embeddings. As illustrated in Fig. 3, we consider two key components of the input meta-path instance pair such as local-neighborhood genetic component-based node attention and topological pair-based component edge weight. Further, the sequential order of nodes and edges is maintained by LSTM cells.

In order to compute the importance of each node in the meta-path instance from comorbidity perspective, it is needed to find the genetic properties of the node which can be acquired from the local neighbors which may consist of different genetic components. Each of these neighbor nodes is associated with multiple genetic instances, however, if they are of the same type, the combined effect of these instances is considered. Therefore, in this work, in order to explain local neighborhood genetic component-based attention, the influence of each neighbor node is first computed before computing the combined effect of the local neighbors.

a) Within genetic component-based node embedding

We compute the influential effect of a genetic component using its local genetic instances. Given genetic component $GC_i = \{G, G-G(PPI), GO, pathway, SNP, miRNA\}$ and node n_k participating in a given meta-path instance, the impact of genetic component GC_i on the node n_k can be found by learning the attention of each of its genetic instances $GC_{ij}, j = 1, 2, .., l$ (local neighbors of node) with respect to the node n_k. For learning the attention of genetic instances, we adopted the idea used by Xiao Wang et al. [51] where the attention of node is influenced by different meta-path based neighbors (end nodes). Thus, the genetic component-based embedding of node n_k can be derived using Eq. (1).

$$h_{n_k}^{GC_i} = \sigma\left(\sum_{(GC_{ij}/n_k)\in GC_i} \alpha^{GC_i}_{n_k,(GC_{ij}/n_k)} . h_{GC_{ij}/n_k}\right) \tag{1}$$

where $h_{n_k}^{GC_i}$ is the learnt node embedding of n_k for the genetic component GC_i, h_{GC_{ij}/n_k} represents the node n_k's local genetic instance GC_{ij}'s embedding of genetic component GC_i and $\alpha^{GC_i}_{n_k,(GC_{ij}/n_k)}$ is the attention coefficient of n_k's local genetic instance GC_{ij} calculated using Eq. (2), which gives the importance of each genetic instance GC_{ij} with respect to n_k.

$$\alpha^{GC_i}_{n_k,(GC_{ij}/n_k)} = \frac{\exp\left(\sigma\left(V^T_{GC_i}\left[h_{n_k}\|h_{GC_{ij}/n_k}\right]\right)\right)}{\sum_{(GC_{il}/n_k)\in GC_i}\exp\left(\sigma\left(V^T_{GC_i}\left[h_{n_k}\|h_{GC_{il}/n_k}\right]\right)\right)} \tag{2}$$

where V_{GC_i} denotes the attention vector of the genetic component GC_i calculated using the multi-head attention mechanism [52]. Each genetic component GC_i comprises of multiple genetic instances $GC_{ij}, j = 1, 2, 3, ..$ embedded with enhanced gene feature representations. In this case, multi-head attention mechanism using linear transformations is applied to all genetic instances GC_{ij}, of genetic component GC_i [52] to derive the attention vector of GC_i.

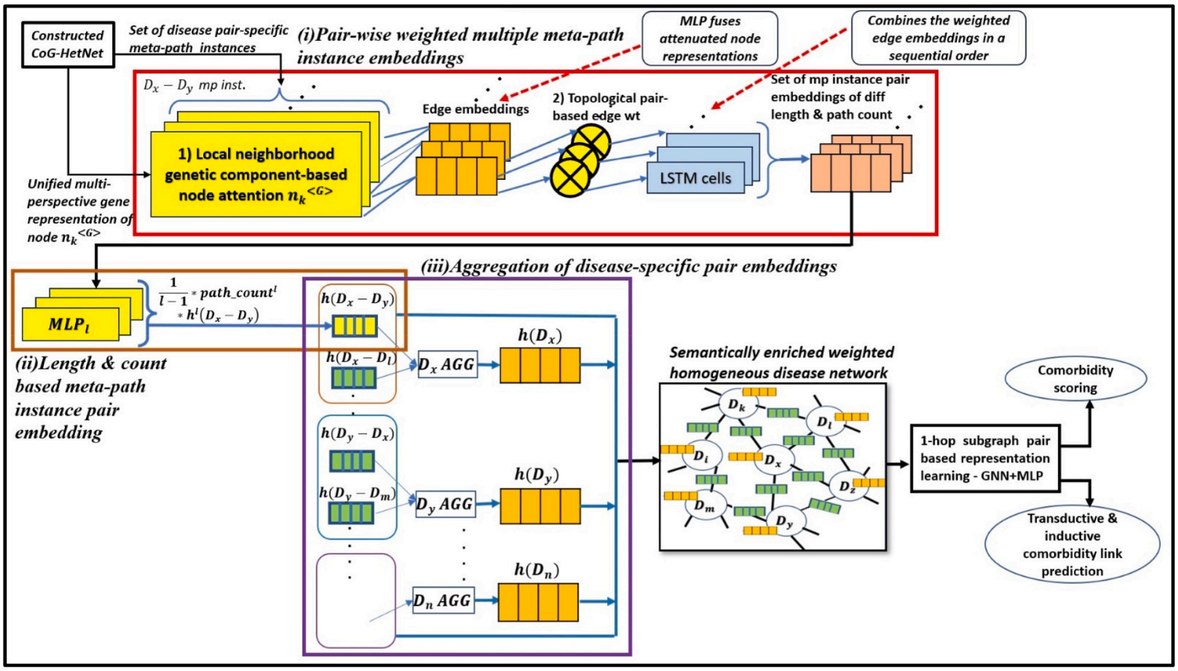

Fig. 3. Genetic semantic-aware weighted meta-path instance embedding based disease node embedding with three main sub-modules- (i) pair-wise weighted multiple meta-path instance embeddings, (ii) length and count based meta-path instance pair embedding and (iii) aggregation of disease-specific pair embeddings.

h_{n_k} and h_{GC_{ij}/n_k} denotes the gene feature representation of node n_k and genetic instance node GC_{ij} with respect to n_k respectively, obtained in section 3.3. σ is the activation function and $\|$ denotes the concatenate operator. $(GC_{il}/n_k) \in GC_i$ in the denominator considers all genetic instances of genetic component GC_i local to n_k.

b) Local neighborhood based node embedding

For a better comprehensive genetic embedding h_{n_k} of each node n_k in the meta-path instance, we need to first learn the attention weight β_{GC_i} of each of its associated genetic component. We then aggregate the attenuated genetic-component specific node embeddings $h_{n_k}^{GC_i}$ using Eq. (3) to embed h_{n_k} with the aggregated influential effect of various genetic components.

$$h_{n_k} = \sum_{i=1}^{n} \beta_{GC_i} . h_{n_k}^{GC_i} \tag{3}$$

where attention weight β_{GC_i} is the normalized importance of all genetic components calculated by softmax function as in Eq. (4).

$$\beta_{GC_i} = \frac{\exp\left(\frac{1}{|V|}\sum_{n_k \in V} \tanh\left(W.h_{n_k}^{GC_i} + b\right)\right)}{\sum_{GC_i \in \{G,GO,pathway,SNP,miRNA,PPI\}} \exp\left(\frac{1}{|V|}\sum_{n_k \in V} \tanh\left(W.h_{n_k}^{GC_i} + b\right)\right)} \tag{4}$$

where V is the set of nodes in the network, W and b are the weight and bias parameters. $h_{n_k}^{GC_i}$ is the genetic component-specific node embedding.

Further, by using equation (3), the final attention weight of a node $\alpha_{n_k}^{<GC_i>}$ can be obtained using another softmax function. The calculated node attention is combined with the unified gene representation of node $n_k{}^{<G>}$ from CoG-HetNet to generate attenuated node representations.

2) Topological pair-based edge weight

Further, a sequence of edge embeddings of the component edges in the given meta-path instance can be obtained by fusing pairs of attenuated node representations using a multi-layer perceptron (MLP) as shown in Fig. 3.

In order to vary the significance of a component edge connected by same pair of nodes shared by multiple meta-path instances, it becomes essential to weigh the component edges based on topologically connected pairs. The component edge weights of the meta-path instance vary with respect to the disease pairs. Hence, we need to find the relative importance of a component edge for a given disease pair among all other pairs. With the availability of sequence of different genetic instances along the comorbid-specific meta-path instance, a topological based similarity score for each component edge along the disease pair-specific meta path instance can be calculated.

For example, the component edge weight w in $D_x \leftrightarrow^{w} GC_{ij}$ for meta-path instance $D_x - GC_{ij} - G_j - D_y$ is calculated as in Eq. (5), by number of $D_x - GC_{ij}$ connections with D_z, $\#(D_x - GC_{ij} - * - D_y)$, given the total degree of $(D_x - GC_{ij})$ and D_z connections represented by $\#(D_x - GC_{ij})$ and $\#(* - D_z)$ respectively and the inverse frequency of $(D_x - GC_{ij} - * - D)$ connecting other diseases. For the calculating the relative importance of $D_x - GC_{ij}$ for D_z, we find the inverse frequency of $D_x - GC_{ij}$ connecting other diseases D, like weighing word in text mining [6]. '*' denotes 0 or any number of genetic instances. A similar edge-weight calculation method is adopted for computing each of the component edge weights for disease pair-based meta-path instances of lengths 2,3 and 4.

$$w = \frac{\#(D_x - GC_{ij} - * - D_y)}{\#(D_x - GC_{ij}) \cup \#(* - D_y)} \left[\frac{1}{\#(D_x - GC_{ij} - * - D)}\right] \tag{5}$$

Once the component edge weights are computed, it is combined with the edge embeddings, discussed earlier in this section. The final weighted component edge embeddings are combined in a sequential order using the LSTM cells similar to Xiang Wang et al. [37]. In this way, a set of multiple weighted meta-path instance embeddings of different lengths and path count is generated for each disease pair. The weight of the meta-path instance pair embedding is further improved by considering both length and count of pair-wise meta-path instances in section 3.3.2.

2.3.2. Length & count based meta-path instance pair embedding

Once the pair-wise multiple meta-path instance embeddings of different lengths and counts generated, a disease pair embedding is produced by combining pair-wise meta-path instance embeddings of length l = 2, 3 and 4 separately. The length of connection greatly affects the semantic strength of the pair, especially comorbidity as discussed in section 1. When there are larger number of shorter and longer paths, then the shorter paths have higher influence on comorbidity than connections with longer paths [53]. Hence, it is important to weigh each set of varying lengths of meta-path instances based on both length and length-wise path count. To achieve this, we used MLP_l for fusing different length-wise set of meta-path instance pair embeddings $h^l(D_x - D_y)$ as shown in Fig. 3. The importance of the connection length l and the number of connections $path\_count^l$ of the given length l for disease pair $D_x - D_y$ is established by weighing each fused set of length-wise pair embeddings using $\frac{1}{l-1} * path\_count^l * h^l(D_x - D_y)$. Finally, the length and count based weighted pair embeddings are concatenated to get the final disease pair embedding $h(D_x - D_y)$.

2.3.3. Aggregation of disease-specific pair embeddings

Finally, to generate an enhanced disease embedding produced by using a disease-specific aggregator that aggregates multiple disease-specific pair embeddings obtained in section 3.3.2 is illustrated in Fig. 3. The aggregation function, AVG is the average aggregation that considers the element-wise average of all disease-wise pair embeddings to produce the final single disease representation is given by Eq. (6).

$$h(D_x) = AVG\left(\{h(D_x - D_i), i = 1, 2, ..n\}\right) \tag{6}$$

where $h(D_x)$ is the final aggregated D_x embedding and $h(D_x - D_i)$ is the pair embedding obtained in section 3.3.2.

2.4. Enhanced comorbidity link prediction and scoring

With the obtained disease node embeddings from section 3.3.3 and the meta-path instance pair embeddings from section 3.3.2, the heterogeneous network is converted to a semantically enriched weighted homogeneous network. The converted homogeneous network is then used for comorbidity link prediction and scoring. In addition, when a new disease node emerges, it is first enriched with the semantic structural knowledge and further added to the transformed homogeneous network for inductive link prediction.

2.4.1. Transforming heterogenous to a semantically enriched weighted homogeneous network

For an enhanced comorbidity link prediction and scoring, it is important to comprehensively learn both direct and indirect relationships between the disease nodes. The direct disease relationships extensively mine the direct comorbidities and the indirect relationships reflect the different semantic aspects between the disease pairs. The direct disease relationships are preserved by first converting the heterogeneous CoG-HetNet to a homogeneous GSWHomoNet by removing the nodes other than disease nodes in CoG-HetNet [40]. This results in a homogeneous network with only connected disease nodes as shown in Fig. 3. The indirect heterogeneous semantics is preserved in the converted GSWHomoNet by encoding the nodes using disease-specific aggregated instance embedding (section 3.3.3) and the edges using pair-wise multiple meta-path instance embeddings (3.3.2) as shown in Fig. 3. The converted network is further trained by 1-hop subgraph based Graph Neural Network (GNN) model for comorbidity scoring and link prediction which is discussed in section 3.4.3. Before this, we first describe the process of enriching the knowledge of unknown disease node which is essential for inductive link prediction.

2.4.2. Enriching the structural knowledge of unknown disease node for inductive link prediction

The comorbidity links can be inductively predicted between known and newly emerging **unknown disease D_{UK}** from outside the network. The D_{UK} node is initially isolated from other disease nodes in the GSWHomoNet. Hence, there is a need to first establish a semantically enriched connection between unknown disease node and known diseases in the GSWHomoNet. This can be achieved by first extracting the unknown diseases from disease ontology [54]. The unknown diseases **D_{UK} s** are extracted using similar ontological relationships of the input known comorbidity pairs. We found that the already existing comorbid diseases are mainly parent-child, sibling, cousin relationships. Hence, we choose the diseases that have similar ontological relationships with the comorbid diseases as they have more chance of being comorbid. In this work, the known 1242 ICD9 diseases were first mapped to DOIDs in the disease ontology and then we were able to extract 562 unknown diseases. We then solve the problem of missing links of the extracted unknown disease nodes D_{UK} in the homogeneous network by extracting D_{UK}'s genetic association information from biomedical data. D_{UK}'s association information is then converted into indirect connections with the known disease nodes in the heterogeneous CoG-HetNet. With the established possible comorbidity links, D_{UK} node is further enriched with genetic and semantic aware structural knowledge using its weighted meta-path instance embedding based disease node embedding as discussed in section 3.3. The enriched D_{UK} node along with its established comorbidity heterogeneous connections are further utilized to bridge direct link with the comorbidity diseases in the homogeneous network GSWHomoNet. Further prediction is carried out using the GNN model discussed in section 3.4.3.

2.4.3. Comorbidity link prediction and scoring

The converted GSWHomoNet obtained in section 3.4.1, is further utilized to learn comorbidity subgraph pair representation for the given disease pair by adopting a similar idea from Jinjiang Juo et al. [55]. The one-hop subgraphs are chosen as we mainly deal with direct comorbidities of a given disease. Though other nodes at different hops are truncated, we are still able to preserve the comorbidity information of the progressing disease nodes using the enriched disease node embedding. Once the one-hop subgraphs $SG^K_{D_i \to D_j}$ and $SG^L_{D_j \to D_i}$ of $D_i - D_j$ pairs are extracted, the subgraph pair $SG^{K\times L}_{D_i, D_j}$ is constructed similar to Jinjiang Juo et al. [55]. For the purpose of training GNN model, a set of subgraph pairs $SG^{K\times L}_{D_i, D_j}$ is labelled with higher or lower chance of being comorbid based on the relative risk scores provided in the standard comorbidity dataset, discussed in section 3.2. Further the subgraph pair representation $h_{SG_{D_i, D_j}}$ is learnt using semantically embedded nodes and edges. An MLP is then used to derive the comorbidity score for already linked known-known (K-K) pairs, predict the links between unlinked K-K pairs (transductive link prediction) and K-UK pairs (inductive link prediction). For finding the comorbid score of already linked K-K comorbid pairs, we give them as test pairs to the trained model by removing their links. Thus, we predict the comorbidity links for both known-known and known-unknown disease pairs.

3. Results and discussion

The main purpose of the evaluation is to determine the effectiveness of the proposed genetic semantic aware weighted homogeneous network embedding method (GSWHomoNet) for transductive and inductive comorbidity link prediction and scoring. Section 4.1 presents the individual and combined effect of the novel modules constituting the GSWHomoNet-based method for comorbidity link prediction and scoring. Section 4.2 shows the effect of GSWHomoNet in comparison with other comorbidity methods for comorbidity transductive link prediction and scoring. This section further presents the evidences (literature, pathways and PPIs) for the newly predicted comorbidity pairs. Finally, a case study is detailed in section 4.3, proving the ability of the proposed GSWHomoNet in discovering the most comorbid disease pairs with evidences which are not found in the standard patient comorbid dataset.

Using five-fold cross validation method, the idea is to split the training dataset into train and validation set while a separate portion of the given dataset is used as a test set to ensure that no data from test set is used during training or validation to prevent data leakage. For evaluation, the dataset is split into training, validation and test set for analysis of prediction performances using area under receiver operating characteristic (AUROC) and area under precision recall (AUPR) scores on the comorbidity dataset. We found that the proposed method of comorbidity link prediction achieved a remarkable performance at RR > 2 than other thresholds when compared to other methods. Hence, we used the RR > 2 as the gold standard for both link prediction and scoring tasks.

3.1. Effect of the novel modules of GSWHomoNet

The effect of each of the novel modules of GSWHomoNet is shown in Table 3 using components A, B, C that evaluates the transductive link prediction and comorbidity scoring and component D focuses on inductive link prediction. For comparison, a network-based approach for comorbidity disease relationships, mpDisNet [19] and its modified variants are used to evaluate the effect of each of the proposed modules successively in Table 3.

3.1.1. Effect of the use of additional genetic components and node features

We will first evaluate the effect of the enhanced comorbidity heterogeneous network CoGHetNet constructed with additional genetic components and enriched node features by carrying out only for transductive comorbidity link prediction and scoring as shown in Table 3. For evaluation, we used mpDisNet [19] which exploited disease-miRNA-gene network is compared with different variants of CoGHetNet such as CoGHetNet-V1 and CoGHetNet-V2 for comorbidity prediction and scoring tasks as described in Table 3. On observation, we find that CoGHetNet V1 and V2 achieved better performance in both comorbidity link prediction and scoring tasks compared to mpDisNet [19] which used only limited genetic components. In addition, we observe that both the variants of CoGHetNetachieved same performances as the meta-path based random walk node embedding strategy could not incorporate the node features to generate node embedding for comorbidity link prediction and scoring. Hence, in order to demonstrate the effect of using node features in CoGHetNet-V2, we need to replace meta-path based random walk by a method that considers node features to learn the final node embedding. Therefore, we show its effect by combining with successive component B as shown in Table 3.

3.1.2. Effect of meta-path based node embedding including node features

We evaluate the effect of considering the deeper semantic information of meta-path instances such as node features, node and edge weights for meta-path instance embedding which are further used to generate the node embedding. For this purpose, we used meta-path based node embedding method, mpDisNet [19] to prove the importance of considering the information enriched in nodes of the meta-path instances. To show the importance of considering the deeper semantic information of both nodes and edges, we compared against MATHNMDA [24] that used only node features in meta-path instance embedding. Thus, the AUROC and AUPR scores of GSWHomoNet-V1 indicate that considering deeper semantic information of meta-path instances have shown a significantly greater impact on the learnt node representations.

As mentioned earlier in component A, the effect of the use of gene enriched node features in the CoGHetNet-V2, can be demonstrated by applying a node embedding strategy that makes use of node features. Hence, we apply the proposed meta-path instance embedding based node embedding method used by the variant GSWHomoNet-V1 onto modified mpDisNet-V1 to show that node embeddings learnt from enhanced heterogeneous network with both enriched node features and additional genetic components play a significant role in comorbidity prediction and scoring. On observation form Table 3B, we find that GSWHomoNet-V1 achieved better results for both prediction and scoring tasks compared to modified mpDisNet-V1. This shows that enhancing the heterogeneous network with more gene-enriched genetic components has a major effect on improving the performance of comorbidity prediction and scoring.

Further, we prove the effect of considering pair-wise length and count of meta-path instances in addition to the proposed meta-path instance embedding to generate the final disease node embedding, by comparing the variants GSWHomoNet-V1 (not considers path length and count) and GSWHomoNet-V2 (considers path length and count). The AUROC and AUPR scores of GSWHomoNet-V1&V2 in Table 3B, indicate that an additional weight contributed by both path length and count significantly improves comorbidity link prediction and scoring.

In addition, we also show the effect of the proposed meta-path instance embedding based node embedding method, GSWHomoNet-V2 by applying it for association prediction tasks across different datasets (disease – microbe using biomedical network dataset and author-author using non-biomedical DBLP dataset) in Table 4. By this, we emphasize the fact that the proposed node embedding method can work across various heterogeneous network datasets. The statistics of heterogeneous network datasets such as disease-microbe and DBLP are given in Refs. [24,56] respectively.

We used Python with Tensorflow in a GPU machine of Intel Xenon with 64 GB RAM to implement the methods with a 5-cross fold validation and repeated the experiments to obtain average AUROC and AUPR scores. We employed Adam optimizer to optimize the performance and adjusted the hyperparameters and found that AUROC and AUPR scores were stable for the values in Table 4. We used the LSTM units of 256 as in Ref. [37] and trained the model for 100 epochs and with an early stopping set with patience of 10.

From the AUROC and AUPR scores in Table 3, we can observe that GSWHomoNet-V2 has achieved a remarkable performance compared to the baseline meta-path-based node embedding methods with the parameter settings in Table 4.

. Thus, the values show that considering deeper semantic information of meta-path instances such as node features, node and edge weights play a vital role in learning node representations that not only achieves better performance in predicting disease-disease comorbidity relationship using CoGHetNet but also other biomedical and non-biomedical heterogeneous network association predictions depicted in Table 5.

3.1.3. Effect of enriched homogeneous network using heterogenous features

To demonstrate the effect of the semantically enriched homogeneous network using heterogeneous features, we used modified mpDisNet-V2 in Table 3C, by adding an additional component of transforming the disease-miRNA-gene heterogeneous network into a disease homogeneous network. The transformed homogeneous network is not semantically enriched and the learning of representations from heterogeneous and homogeneous networks are independent as in Ref. [56] which are then fused to generate the final disease node embedding. In addition, we also compared the variants of GSWHomoNet V3 and V4 using different forms of homogeneous network without and with semantic enrichment, respectively. The AUROC and AUPR scores in Table 3C, shows that GSWHomoNet-V3 and V4 achieved better performance for comorbidity prediction and scoring compared to modified mpDisNet-V2 without sematic enrichment to the converted homogeneous network. This shows the effect of learning rich node representation using heterogeneous network information by both GSWHomoNet-V3 and V4 for comorbidity prediction and scoring.

However, a semantically enriched homogeneous network

Table 3
Effect of the novel components and their combined effect on comorbidity link prediction and comorbidity scoring.

S.No	Methods	Remarks	Comorbidity link prediction						Comorbidity scoring
			Training		Validation		Testing		
			AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	Correlation
A	**Effect of the use of additional genetic components and node features**								
I.	mpDisNet [19]	1. Uses two genetic components – **gene and miRNA** **2. No node features** **3. Node embedding strategy- meta-path based random walk**	0.641	0.644	0.640	0.642	0.644	0.652	0.632
II.	Biswas et al. [22]	1. Uses genetic components – genes, PPI, GO and pathway 2. Node features embedded as vector of complex numbers **3. Node embedding strategy- meta-path based random walk**	0.668	0.665	0.664	0.667	0.669	0.671	0.653
III.	**CoGHetNet-V1**	1. Uses genetic components – gene, GO, pathway, PPI, miRNA, SNP 2. No node features **3. Node embedding strategy- meta-path based random walk**	**0.689**	**0.686**	**0.688**	**0.690**	**0.691**	**0.696**	**0.711**
IV.	**CoGHetNet-V2**	Same as method-III but node features- enriched gene feature representation	**0.687**	**0.689**	**0.690**	**0.685**	**0.691**	**0.696**	**0.711**
B.	**Effect of meta-path based node embedding including node features**								
V.	mpDisNet [19]	Applied meta-path based random walk to generate disease node embedding	0.641	0.644	0.640	0.642	0.644	0.652	0.632
VI.	MATHNMDA [24]	Meta-path instance embedding is based on the *aggregation* of node features from **CoGHetNet-V2 -** further utilized to generate disease node embedding	0.732	0.729	0.728	0.726	0.731	0.729	0.737
VII.	GraphMSE [36]	Meta-path instance embedding is based on the *concatenation* of node features from **CoGHetNet-V2 -** further utilized to generate disease node embedding	0.740	0.737	0.738	0.735	0.742	0.738	0.746
VIII.	**GSWHomoNet-V1- with CoGHetNet-V2**	Meta-path instance embedding incorporated node features, node and edge importance to generate pair embedding- further combined using MLP to generate final disease node embedding	0.853	0.855	0.849	0.851	0.852	0.856	0.784
IX.	Modified mpDisNet-V1	Applied our variant method VIII after enriching all nodes in disease-gene-miRNA network with gene features.	0.797	0.793	0.789	0.792	0.796	0.788	0.625
X.	**GSWHomoNet-V2- with CoGHetNet-V2**	Adopted the meta-path instance embedding method used by method VIII - further weighted using path length and count by MLP to generate the final disease node embedding	**0.878**	**0.882**	**0.874**	**0.877**	**0.876**	**0.881**	**0.793**
C.	**Effect Of Enriched Homogeneous Network Using Heterogenous Features**								
XI. XI.	Modified mpDisNet-V2	1. Transformed the heterogeneous disease-miRNA-gene network into homogeneous disease network without semantic enrichment 2. Fused the indirect relationship learnt using meta-path based random walk from heterogeneous network and direct relationship from homogeneous network as in Ref. [40] to generate final disease node embedding	0.669	0.674	0.664	0.668	0.667	0.671	0.640
XII. XII.	**GSWHomoNet-V3**-with **CoGHetNet-V2**	1. Transformed the heterogeneous CoGHetNet-V2 into homogeneous disease network CoGHomoNet without semantic enrichment 2. Fused the indirect relationship learnt using method X from CoGHetNet-V2 and direct relationship from homogeneous network CoGHomoNet as in Ref. [40] to generate final disease node embedding	**0.876**	**0.884**	**0.880**	**0.882**	**0.872**	**0.895**	**0.806**
XIII. XIII.	**GSWHomoNet-V4 – withCoGHetNet-V2**	1. Transformed CoGHetNet-V2 into homogeneous disease network CoGHomoNet with semantic enrichment 2. Learnt both indirect and direct semantics from a single CoGHomoNet using GNN + MLP discussed in section 3.4.	**0.894**	**0.899**	**0.891**	**0.893**	**0.895**	**0.903**	**0.848**
D.	**Effect of Use of Semantically Rich Structural Information for Inductive Link Prediction**								
XIV. XIV.	Modified mpDisNet-V3	1. Adopted method V – learnt only heterogeneous network information using meta-path based random walk (No homogeneous conversion) 2. Supplied unknown disease node with similarity based structural information [47]	0.622	0.629	0.622	0.624	0.621	0.626	–
XV. XV.	Modified mpDisNet-V4	Same as method XIV but supplied unknown disease node with semantically rich structural information	0.652	0.650	0.647	0.645	0.649	0.642	
XVI. XVI.	**GSWHomoNet-V5**-(uses GSWHomoNet-V4)	1. Adopted method XIII – learnt both indirect and direct semantics from converted homogeneous network. 2. Supplied unknown disease node with similarity based structural information [47]	**0.804**	**0.806**	**0.798**	**0.803**	**0.800**	**0.809**	–
XVII.	**GSWHomoNet-V6**-(uses GSWHomoNet-V4)	Same as method XVI but supplied unknown disease node with semantically rich structural information	**0.862**	**0.870**	**0.861**	**0.869**	**0.860**	**0.873**	–

Table 4
Optimal parameter settings of the methods for different datasets.

Dataset	Learning rate	L2 regularization	Dropout	Embedding size
Biomedical datasets (Disease-microbe & CoGHetNet)	0.005	0.001	0.5	128
Non-biomedical dataset (DBLP)	0.002	0.001	0.6	128

Table 5
Association prediction performance evaluation of node embeddings generated by different meta-path instance embedding based methods across different datasets.

Meta-path based node embedding methods	Disease-microbe (D-m)		DBLP (A-A)	
	AUROC	AUPR	AUROC	AUPR
mpDisNet [19]	0.732	0.735	0.711	0.713
MATHNMDA [24]	0.807	0.812	0.741	0.738
GraphMSE [36]	0.811	0.814	0.746	0.742
GSWHomoNet V1	0.826	0.828	0.832	0.835
GSWHomoNet V2	0.834	0.839	0.854	0.857

established by GSWHomoNet-V4, have shown a significant impact on both prediction and correlation (strength and direction of linear relationship between two variables for n samples and ranges from −1 to +1) performances compared to GSWHomoNet-V3. The main reason is that the indirect semantics encoded in both nodes and edges in the converted homogeneous network helps to update the direct comorbidity context with the indirect semantics from a single homogeneous network itself.

3.1.4. Effect of use of semantically rich structural information for inductive link prediction

For inductive link prediction, we adopt the similar experimental settings as in Ref. [45], by hiding 10 % of nodes, the remaining set of nodes and edges are used for training while the hidden nodes were used for testing. To show the effect of the semantically rich structural knowledge supplied to the unknown disease node, we used mpDisNet [19] and attached an additional step of supplying knowledge to the unknown disease node shown in Table 3D using the variants of mpDisNet [19] such as Modified mpDisNet-V3 and V4. However, the knowledge supplied to the unknown disease node plays a crucial part in predicting comorbidity links between known and unknown disease nodes. Therefore, we used the variants modified mpDisNet-V3, V4 and GSWHomoNet-V5, V6 by supplying unknown disease node with only similarity based structural information [45] and semantic structural information. On observation in Table 3D, we find within the variants of modified mpDisNet and GSWHomoNet, modified mpDisNet-V4 and GSWHomoNet-V6 gives better prediction results compared to modified mpDisNet-V3 and GSWHomoNet-V5 respectively. This may be because supplying only node's attributes based structural similarity information might result in missing the links between nodes that may be semantically connected. Therefore, encoding the unknown node with semantically rich structural information with other known nodes help improve predicting the comorbidity links inductively. Thus, GSWHomoNet-V6 method which has produced better results in comorbidity scoring and prediction tasks is used for further evaluation in sections 4.2 and 4.3.

3.2. Evaluation of comorbidity-based methods for transductive link prediction and comorbidity scoring

Among different variants of the GSWHomoNet in Table 3, we find GSWHomoNet V4 has achieved better prediction as well as significant correlation results compared to other variants for transductive link prediction and comorbidity scoring. Hence, we use GSWHomoNet V4 for further evaluation with state-of-art comorbidity methods including both statistical and network-based approaches illustrated in Fig. 4. On observation, the proposed method outperforms other state-of-art methods covering larger area under both ROC and PR curves. This is because GSWHomoNet V4 learns both direct and richer indirect semantics of the comorbidity disease pairs effectively while the other comorbidity-based methods do not effectively incorporate richer semantic comorbidity pair information.

This demonstrates the effectiveness of the comorbidity scores obtained by **GSWHomoNet V4** by correlating the scores calculated by various comorbidity-based methods and the normalized relative risk (RR) scores from standard comorbidity dataset as shown in Table 6. In addition, Table 6 shows the evaluated result of the computed scores against the comorbidity scores calculated across different perspectives of comorbidity criteria such as gene-based [2], pathway-based [2,17] and PPI-based [17]. Using the sources in Table 1, we obtain the disease associated genes, pathways and PPIs between comorbidity pairs. Then we utilize the extracted association information to calculate the comorbidity scores from different comorbidity perspectives such as genes, pathways and PPIs using the Jaccard index as in Ref. [13].

On observation it is found that the comorbidity scores obtained by GSWHomoNet V4 gives a remarkable correlation compared to other methods for relative risk scores in the standard patient comorbidity dataset. In addition, the method achieves a significant correlation in the PPI aspect of comorbidities. However, for gene-based, GSWHomoNet V4 achieves relatively good correlation but less than XD score [57], ComoR-Gene [2] and Genetic-functional measure [16] and less than ComoR-Pathway [2] for pathway-based. This is because, these baseline methods have directly used only shared genes or shared pathways to derive the final comorbidity scores and for other aspects, these methods do show a significant correlation result, as they are biased by a single genetic component. However, GSWHomoNet V4 considers the contribution of multiple genetic components whose combined effect has significant impact on the standard patient comorbidity dataset. Overall, GSWHomoNet-V4 achieves better correlated results not only with standard relative risk scores of the patient dataset but also with different perspectives of comorbidity. This proves that enriching sufficient semantic knowledge of more genetic components helps in incorporating deeper comorbidity context which is reflected by the comorbidity scores. On the other hand, GSWHomoNet-V4 when compared within itself among different aspects of comorbidity, produced better results mainly for PPI-based.

The better performing GSWHomoNet-V4 and GSWHomoNet-V6 predicted 1,95,192 and 1,32,356 comorbidity pairs using transductive and inductive link prediction respectively. To validate the discovered comorbidity pairs, we provided literature evidence, for instance, the disease pair with no comorbid link in the standard comorbid dataset [1], "dementia-amnestic disorder" is evidenced in PMID: 7854546, stating that the study determined "cholinergic blockade fully reproduces the amnestic disorder found in dementia", "diabetic angiopathy-acute myocardial infraction" in PMID: 8711913 talks about diabetic angiopathy in pregnant women, later stated "The occurrence of acute myocardial infarction in pregnant diabetic women is associated with elevated fetal and maternal mortality." Similarly, for inductively predicted pairs with one of the diseases available in standard dataset, the pair "systemic lupus erythematosus- Mikulicz disease" in PMID: 576964, is inferred to co-occur in the sentence "Mikulicz disease and subsequent lupus erythematosus development.", another pair "Type 2 diabetes mellitus-female breast cancer" in PMID: 27509959 clearly states the cause of female breast cancer using the sentence "Type 2 Diabetes Mellitus as a Risk Factor for Female Breast Cancer in the Population of Northern Pakistan." Further we analyzed the comorbidity relationships for those pairs with support of literature evidence by using the shared pathways and PPI patterns as in Ref. [22]. The shared pathways are obtained by mapping of common genes to pathways as in Ref. [59]. The shared PPIs are obtained by using STITCH [59] with the disease

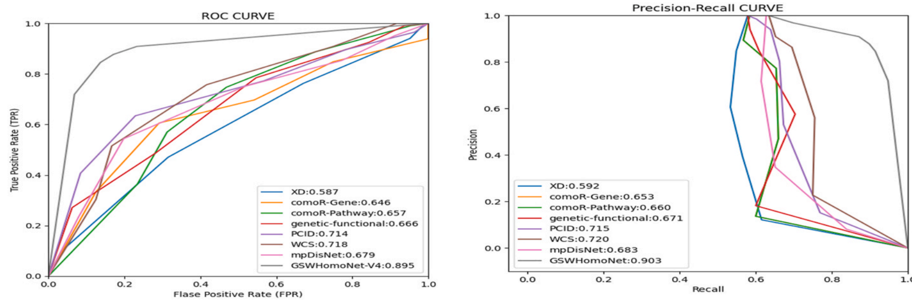

Fig. 4. AUROC and AUPR for transductive link prediction performance comparison of various comorbidity-based methods.

Table 6
Spearman's rank correlation coefficient between different perspectives of comorbidity scores and computed scores by various comorbidity-based methods for comorbidity pairs from standard comorbidity dataset.

Comorbidity-based scoring methods	Spearman's rank correlation			
	RR [1]	Gene-based	Pathway-based	PPI-based
Statistical comorbidity scoring				
XD score [57]	0.264	0.912	0.206	0.254
comoR-Gene [2]	0.577	0.895	0.235	0.421
comoR-Pathway [2]	0.506	0.219	0.887	0.268
Genetic-functional measure [7]	0.584	0.954	0.215	0.316
PCID [17]	0.7165	0.826	0.723	0.844
WCS [58]	0.743	0.834	0.729	0.847
Network embedding-based comorbidity scoring				
mpDisNet [19]	0.632	0.859	0.532	0.814
GSWHomoNet V4	0.848	0.873	0.756	0.857

associated genes.

Thus, some unlinked diseases in GSWHomoNet were transductively predicted to be comorbid pairs. The additional comorbid pairs were discovered inductively between the unknown diseases outside the network and the diseases already in GSWHomoNet. These predictions are validated against three different evidences shown in Table 7 such as literature evidence using PMID, the existence of pathways mapped via common genes between the diseases and common PPI patterns similar to Biswas et al. [22]. Further, the discovered comorbid pairs that hold all three evidences, strongly proves that they have more chance of being comorbid. The predicted comorbid disease pairs against the standard comorbid dataset having all three evidences are listed in Table 7.

In addition, it is necessary to investigate why certain links are predicted between diseases by GSWHomoNet-V4 (transductive) and GSWHomoNet-V6 (inductive) that utilizes GNN. As already discussed, certain links between diseases are induced by GNN model trained on a set of disease pair subgraphs. For the given target disease nodes, the trained model can generate disease subgraph pair representation embedded with genetic meta-path information of various disease nodes of the input subgraphs. However, it is important to uncover the most influencing meta-path information contributing to the subgraph representation for comorbid link prediction. For this, integrated gradient technique can be applied that takes the generated subgraph pair representation of the given target disease nodes, back into GNN and calculate partial derivatives of the predicted scores which can be integrated to get the overall integrated gradients for the elements in the final subgraph representation (final layer). The highest integrated gradient score in the final layer's representation gives the most influential meta-path information for the model's prediction. For example, the prediction of comorbid link between Cardiovascular disease (CVD) and Type 2 Diabetes (T2D) is analyzed using integrated gradient. The meta-path "CVD-IRS1-Beta_cell function-CDKN2B-T2D" (Disease-Gene-Pathway-Gene-Disease) is found to be the most contributing factor to this comorbid link prediction. Further, all the genetic components in this metapath are supported in the published literature such as "IRS1" and "CDKN2B″ in PMID:29938349 and "Beta-cell function" in PMID:28957322, thus confirming the genetic comorbid connection between Cardiovascular disease and Type 2 Diabetes.

3.3. Case study

To illustrate the capability of GSWhomoNet-based method in discovering new comorbid disease pairs, we selected the most common index diseases such as alzheimer's disease and cardiovascular disease that have higher probability of giving rise to comorbidities and carried out the case study using the standard comorbid patient dataset mentioned in section 3.2. In this study, for each index disease, we presented the top 5 comorbid pairs in Table 8 with the availability information in the dataset. The availability information is categorized into three: 1) Both the diseases D1 and D2 are available as a comorbid pair in the dataset, 2) D1 and D2 are available without comorbid link in the comorbid dataset and 3) either D1 or D2 is alone available in the dataset. For the newly identified comorbid pairs, the corresponding literature evidence is provided. On observation, we find that most of the top ranked comorbid pairs are newly discovered ones. Thus, by employing the proposed method, we found additional newly discovered comorbid pairs that are not in standard comorbid dataset. To validate the newly discovered comorbid pairs, the corresponding literature evidence is provided as shown in Table 8. On observation, we find that most of the top ranked comorbid pairs are newly discovered ones with support of literature evidence. This implies that a richer genetic aspect is essential to establish a strong comorbid link. The additionally discovered genetic-based comorbidity pairs can be added to the patient-based comorbid dataset to increase the number of comorbid pairs as well as help find solutions to avoid risks of the patients from progressing further to other comorbid diseases.

4. Conclusion

For effective comorbidity link prediction and scoring, learning both rich indirect and direct semantics between comorbidity pairs is essential. To solve this, we first constructed an enhanced gene enriched CoGHetNet with several genetic components and employed a weighted meta-path instance pair embedding that learnt a semantically rich node embedding. Further, we used both disease pair embedding and disease node embedding to encode the transformed homogeneous disease network forming a genetic-semantic aware weighted homogeneous network, GSWHomoNet. Thus, we attempted to learn both indirect and direct enriched comorbidity contexts for enhanced comorbidity link prediction and scoring. With such an enriched semantics, GSWHomoNet-based method achieved better accuracy for each of its proposed components as well as their combined effect on both comorbidity link prediction and scoring tasks compared to other comorbidity-based methods. Neverthless, we also discovered comorbidity disease pairs both transductively and inductively for which some pairs with

Table 7
Predicted comorbid disease pairs in comparison with the standard comorbid dataset with support of all three evidences.

	Disease 1	Disease 2	Literature evidence (PMID)	Sharing pathways mapped via common genes	Sharing PPI patterns
Disease 1 and 2 appear in standard comorbid dataset [1] without comorbid link (Transductive link prediction)	dementia	amnestic disorder	7854546	Neuroactive ligand-receptor interaction, MAPK signaling pathway	MAPT-BDNF, APP-BDNF, APOE-BDNF
	Cardiovascular disease	Non-alcoholic fatty liver disease	36965839	p53 signaling pathway, Regulation of TLR by endogenous ligand	KCNQ1-INS, ACE-ADIPOQ
	diabetic angiopathy	acute myocardial infraction	8711913	Cell surface interactions at the vascular wall, MicroRNAs in cancer	FN1-PLAT, CAT-MB
	esophageal cancer	acute myeloid leukemia	33744998	Transcriptional misregulation in cancer, DAP12 interactions	RNF6-BRPF1
	major depressive disorder	pneumonia	34801384	mTOR signaling pathway, Creation of C4 and C2 activators	NOTCH3-TNF, NOTCH3-TWIST1
	oral cavity cancer	esophageal cancer	19843324	Proteoglycans in cancer, MicroRNAs in cancer	RRM2-TP53P1, VMP1-ZEB1
	membranous glomerulonephritis	erythema multiforme	6774668	IL-17 signaling pathway, NF-kappa B signaling pathway	RAC3-TNF, TSLP-IFNG
	diarrhea	cerebral artery occlusion	33752418	Pathogenic Escherichia coli infection, NOD-like receptor signaling pathway	FOXP3-IL6
	hyperthyroidism	major depressive disorder	23438714	Gastrin-CREB signalling pathway via PKC and MAPK, Signaling by GPCR	TG-FKBP5, TRH-HTR2A
	major depressive disorder	peritonitis	34309667	T cell receptor signaling pathway, Platelet degranulation	TPH2-IL6
One of the diseases alone is available in standard comorbid dataset [1] (Inductive link prediction)	systemic lupus erythematosus	Mikulicz disease	576964	Cytokine Signaling in Immune system, NF-kappa B signaling pathway	TRIM21-ELF2, TRIM21-UNC93B1
	Urinary bladder cancer (D001749)	breast cancer (D001943)	32845432	E2F mediated regulation of DNA replication, Inhibition of replication initiation of damaged DNA by RB1/E2F1	RB1-BARD1
	coronary artery disease (D003324)	peripheral vascular disease (D016491)	36584730	AGE-RAGE signaling pathway in diabetic complications, Metabolism of lipids and lipoproteins	LMF2-VWF, LMF2-SERPINE1
	vascular disease	diabetic angiopathy	11582946	AGE-RAGE signaling pathway in diabetic complications, AMPK signaling pathway	APOB-EDN1
	Chronic myeloid leukemia	invasive ductal carcinoma	28690659	Constitutive Signaling by Aberrant PI3K in Cancer, AMPK signaling pathway	BLK-EGFR, CUL3-EGFR
	Type 2 diabetes mellitus	female breast cancer	27509959	Activation of BH3-only proteins, AGE-RAGE signaling pathway in diabetic complications	HNF1A-EGFR, PAX4-AR
	skin disease	SAPHO syndrome	12719925	IL-17 signaling pathway, activated TAK1 mediates p38 MAPK activation	KRT14-NOD2, ELF3-NOD2
	Coronary artery disease	diabetic angiopathy	8711913	HIF-1 signaling pathway, HDL-mediated lipid transport	APCO3-INS, LPL-INS
	essential hypertension	peripheral artery disease	23126350	Peptide hormone metabolism, MET activates PTK2 signaling	AGT-ACE, NOS3-ACE, AGT-P2RY12
	Major depressive disorder	dissociative disorder	19042786	Serotonergic synapse, Signaling by GPCR	HTR2A-SLC6A4, TPH2-SLC6A4, TPH2-FKBP5

Table 8
Top 5 predicted comorbid disease pairs by GSWHomoNet-based method for Alzheimer's disease and cardiovascular disease.

Index disease D1	Comorbid Disease D2	Availability of D1, D2 in standard comorbid dataset			Literature evidence (PMID)
		D1-D2 with comorbid link	D1 and D2 unlinked	Either D1 or D2 alone	
Alzheimer's disease	Mild cognitive impairment	–	–	Y	15753419
	Lewy body dementia	–	–	Y	37646002
	Explosive personality disorder	Y	–	–	–
	Anosognosia	–	–	Y	235974
	Hyperlipidaemia	–	Y	–	32130251
Cardiovascular disease	Cerebrovascular disease	Y	–	–	–
	Type 2 Diabetes		Y		33648384
	COVID-19	–	–	Y	32845926
	Heart Failure	Y	–	–	–
	Peptic ulcer disease	–	–	Y	29509757
	Renal Dysfunction	–	–	Y	18191240

literature, pathway and PPI evidence have been reported. In addition, we correlated the comorbidity scores obtained against different comorbidity perspectives and found a better correlation of 0.848 in patient comorbidity dataset and 0.857 mainly in PPI-aspect. As the quantified and discovered comorbid pairs share similar genetic characteristics, then the same drug can be used to treat such diseases. For example, cyclosporine drug is repurposed for comorbid conditions like psoriasis and rheumatoid arthritis as shared genetic characteristics between the diseases remain targeted indications for the drug cyclosporine [60]. The limitation of the work is that the intricate details of the biological components like the structural details of pathways and integration of additional comorbidity details like the causal direction of disease occurrence are not included. As future work, an additional enhancement to the comorbidity task would consist of diving into the knowledge embedded in the biological components. In addition, considering patient-specific data such as patient's history, symptoms, treatments, etc. can discover comorbid conditions in patients. Further utilization of history of patient records can uncover the temporal progression of a sequence of comorbid diseases. Such an integration of various biological components with patient-specific characteristics can provide advances for problems in personalised medicine for comorbidities.

CRediT authorship contribution statement

Karpaga Priyaa Kartheeswaran: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Arockia Xavier Annie Rayan:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Conceptualization. **Geetha Thekkumpurath Varrieth:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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